

QUIMPER BRETAGNE AP, France

WMO: **072010**

Lat: **47.9728N** Lon: **4.1606W** Elev: **297** StdP: **14.54** Time Zone: **1.00 (EUC)** Period: **99-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	29.5	31.8	22.3	16.9	33.0	25.3	19.6	34.6	29.8	49.3	27.3	49.8	6.5	40	0.561	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	14.3	80.6	66.9	76.9	65.0	73.7	63.6	68.4	77.6	66.5	73.9	65.0	70.8	8.1	50	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	65.4	95.0	70.8	64.0	90.2	68.7	62.8	86.5	67.2	32.7	78.2	31.2	74.1	30.1	70.7	76.8	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	24.6	21.6	19.2	DB	25.4	86.0	3.2	4.4	23.1	89.2	21.2	91.7	19.4	94.2	17.0	97.3	(4)
(5)				WB	23.7	72.0	3.2	2.4	21.4	73.7	19.5	75.1	17.7	76.5	15.4	78.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	53.5	44.5	44.6	47.5	50.7	55.4	60.7	63.5	63.0	60.3	55.4	49.4	45.9	(6)
(7)			DBStd	8.38	5.31	5.10	4.78	4.83	4.79	4.75	4.33	3.81	4.16	4.49	4.63	5.60	(7)
(8)			HDD50	718	179	159	104	45	8	0	0	0	0	9	65	149	(8)
(9)			HDD65	4303	635	572	542	428	299	147	82	85	151	299	469	594	(9)
(10)			CDD50	1979	9	7	27	68	176	320	417	404	309	175	46	21	(10)
(11)			CDD65	89	0	0	0	0	2	18	35	24	10	0	0	0	(11)
(12)			CDH74	674	0	0	0	4	28	157	282	149	53	1	0	0	(12)
(13)			CDH80	127	0	0	0	0	2	30	54	37	4	0	0	0	(13)
(14)	Wind		WSAvg	9.1	10.1	10.2	10.2	9.7	9.1	8.5	8.3	7.7	7.6	8.2	9.4	10.2	(14)
(15)	Precipitation		PrecAvg	38.0	4.4	3.7	2.6	2.9	2.6	2.0	2.2	2.2	2.4	3.9	4.6	4.4	(15)
(16)			PrecMax	47.1	9.7	8.9	8.1	7.3	6.5	4.8	5.9	4.4	6.1	5.9	9.5	8.6	(16)
(17)			PrecMin	28.1	1.0	0.7	0.3	0.1	0.2	0.2	0.5	0.2	0.1	0.7	0.8	1.0	(17)
(18)			PrecStd	5.2	2.3	2.0	1.7	1.8	1.4	1.2	1.2	1.2	1.6	1.4	2.2	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		55.6	58.5	67.7	73.3	77.4	84.2	85.4	85.4	79.3	69.1	61.8	58.0	(19)
(20)			MCWB		53.6	52.3	53.1	59.4	64.1	70.4	68.0	70.1	66.2	62.7	58.5	55.1	(20)
(21)		2%	DB		54.6	54.4	60.8	67.1	72.4	78.2	80.7	78.4	74.9	66.0	59.7	56.4	(21)
(22)			MCWB		52.8	51.1	52.0	55.8	61.4	64.9	66.8	66.8	64.6	60.9	57.4	54.5	(22)
(23)		5%	DB		53.6	53.2	56.7	62.8	68.0	74.4	76.9	73.6	71.5	63.9	58.2	55.3	(23)
(24)			MCWB		52.0	50.7	50.6	53.9	58.8	63.6	65.3	64.4	62.8	60.0	55.9	53.7	(24)
(25)		10%	DB		52.5	52.0	54.2	58.7	64.1	70.6	72.6	70.3	68.1	62.3	56.7	54.1	(25)
(26)			MCWB		50.8	49.8	50.5	52.1	57.2	61.6	63.3	63.0	61.4	59.2	54.5	52.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		54.8	53.9	55.2	60.7	65.6	70.8	70.3	71.9	69.0	64.3	60.5	56.4	(27)
(28)			MCDB		55.1	55.8	61.5	69.1	74.6	81.2	81.7	83.6	76.9	67.5	60.8	56.9	(28)
(29)		2%	WB		53.7	52.6	53.6	57.3	62.3	67.1	67.9	67.8	66.2	62.6	58.0	55.2	(29)
(30)			MCDB		54.4	53.9	56.8	64.7	69.9	75.9	78.5	75.0	71.6	64.4	58.9	56.1	(30)
(31)		5%	WB		52.5	51.5	52.6	54.8	60.2	64.5	66.0	65.8	64.5	61.3	56.5	54.1	(31)
(32)			MCDB		53.4	52.7	55.3	60.2	66.6	72.5	74.2	71.0	68.4	63.0	57.8	55.2	(32)
(33)		10%	WB		51.2	50.3	51.4	53.2	58.0	62.3	64.4	64.3	63.0	60.0	54.8	52.8	(33)
(34)			MCDB		52.3	51.8	53.9	57.4	62.7	68.6	70.6	68.4	66.5	61.7	56.4	54.1	(34)
(35)	Mean Daily Temperature Range		MDBR		9.6	10.6	12.2	14.1	14.4	14.9	14.3	14.3	14.7	11.9	10.6	9.4	(35)
(36)		5% DB	MCDBR		8.9	10.6	16.4	20.4	20.3	22.3	22.7	19.6	19.1	13.4	11.1	7.8	(36)
(37)			MCWBR		8.5	8.5	10.0	11.4	11.6	12.1	11.8	10.2	10.5	8.9	9.2	7.8	(37)
(38)		5% WB	MCDBR		8.3	8.9	12.3	16.5	18.1	20.4	19.7	16.3	15.2	10.8	10.0	7.7	(38)
(39)			MCWBR		8.6	8.4	8.7	10.0	10.9	11.9	11.0	9.6	9.6	8.4	9.6	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.333	0.347	0.371	0.381	0.385	0.380	0.379	0.369	0.357	0.356	0.336	0.327	(40)
(41)		taud			2.402	2.384	2.333	2.322	2.356	2.394	2.400	2.448	2.462	2.455	2.428	2.374	(41)
(42)		Ebn at Noon			232	256	267	276	279	280	278	277	270	250	229	217	(42)
(43)		Edh at Noon			22	28	34	37	37	36	36	33	29	25	21	20	(43)
(44)	All-Sky Solar Radiation	RadAvg			320	590	969	1496	1766	1920	1834	1579	1260	734	402	271	(44)
(45)		RadStd			26	57	101	129	142	197	91	120	89	56	31	30	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page